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Surgical platform system

Abstract

A surgical platform system facilitating manipulation of a patient support thereby is provided. The surgical platform system can include a support supporting a linkage portion, a first platform portion, and a second platform portion relative to the ground. The linkage portion can include a base pivotally attached to the support, a first connector pivotally supported by the base, and a first support portion and a second support portion pivotally supported by the first connector. The first platform portion can be supported by the first support portion and the second support portion, and the second platform can be supported by the base. Pivotal movement of the first platform portion relative to the base via pivotal movement of the connector, and pivotal movement of the second platform portion relative to the support via pivotal movement of the base can serve to separately articulate the first platform portion and the second platform portion relative to one another.

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Background/Summary

FIELD

(1) The present technology generally relates to a surgical platform system having a first platform portion and a second platform portion that can be used to adjust portions of a patient supported thereby before, during, and after surgery in relation to a surgical robotic system.

BACKGROUND

(2) Typically, conventional surgical tables include some form of patient articulation, but such patient articulation afforded thereby is often quite limited. For example, sometimes the conventional surgical tables can afford a limited degree of flexion or extension of the spine of the patient by lifting a portion of the torso of the patient in a upward direction and a downward direction. The patient articulation afforded by the convention surgical tables is limited because portions thereof supporting portions of the patient are typically not separately articulatable to any great degree. For example, portions of the conventional surgical tables supporting the heads and the torsos of the patient and other portions of the conventional surgical tables supporting the legs of the patient are not typically separately articulatable relative to one another in a fashion that affords any

great degree of flexion and extension via such articulation. Therefore, in order to enhance patient articulation, there is a need for a surgical platform system including a first platform portion and a second platform portion that are at least in part separately articulatable relative to one another to afford separate articulation of a first portion of the patient supported by the first platform portion and a second portion of the patient supported by the second platform portion. Such a surgical platform system incorporating the first platform portion and the second platform portion separately articulatable relative to one another can correspondingly position/orient and reposition/reorient a first portion of the patient's body supported by the first platform portion, and a second portion of the patient's body supported by the second platform portion with respect to one another. Portions of the surgical platform system can be attached relative to and/or integrated with a surgical robotic system to afford positioning and repositioning the patient's body relative thereto before, during, and after surgery.

SUMMARY

(3) The techniques of this disclosure generally relate to a surgical platform system that can be used as a surgical table for performing surgery on a patient supported thereby, with a first platform portion and a second platform portion of the surgical platform system capable of supporting a first portion and a second portion, respectively, of the patient thereon. The first platform portion and the second platform portion can be separately articulatable relative to one another, and portions of the surgical platform system can be integrated with a surgical robotic system to afford positioning/orienting and repositioning/reorienting the patient's body relative to the surgical robot before, during, and after surgery.

(4) In one aspect, the present disclosure provides a surgical platform system facilitating manipulation of a patient supported thereby, the surgical platform system including a support portion including a horizontally-oriented portion and a vertically-oriented portion, the horizontally-oriented portion having a first end, an opposite second end, a first end portion at the first end, a second end portion at the second end, and a cross member extending between the first end portion and the second end portion, and the vertically-oriented portion including a column portion supported by the horizontally-oriented portion, the column portion spacing a first platform portion, a second platform portion, and a linkage portion of the surgical platform system from the horizontally-oriented portion, the linkage portion including a first base portion positioned on a first side of the column portion, a second base portion positioned on a second side of the column portion, a first connector extending between the first base portion and the second base portion, and a first support portion and a second support portion supported at least in part by the first connector, the first base portion and the second base portion being pivotally attached relative to the column portion, and the second platform portion being supported by the first base portion and the second base portion, the first connector having a first end and an opposite second end, the first end of the first connector being pivotally attached relative to the first base portion and the second base portion, and the first support portion and the second support portion each including a first end and an opposite second end, the first support portion and the second support portion being attached adjacent the second ends thereof to the second end of the first connector, and the first platform portion being supported by the linkage portion adjacent the first ends of the first support portion and the second support portion; the first platform portion including first patient support portions for supporting portions of the patient's head and upper torso; and the second platform portion including second patient support portions for supporting portions of the patient's legs; where the first platform portion is pivotal relative to the first base portion and the second base portion via pivotal movement of the first connector; where the second platform portion is pivotal relative to the column portion via pivotal movement of the first base portion and the second base portion; and where the pivotal movement of the first platform portion relative to the first base portion and the second base portion, and pivotal movement the second platform portion relative to the column portion serves to separately articulate the first platform portion and the second platform portion

relative to one another.

(5) In another aspect, the present disclosure provides a surgical platform system facilitating manipulation of a patient supported thereby, the surgical platform system including a horizontally-oriented portion and a vertically-oriented portion, the horizontally-oriented portion supporting the vertically-oriented portion, and the vertically-oriented portion spacing a first platform portion, a second platform portion, and a linkage portion of the surgical platform system from the horizontally-oriented portion, the linkage portion including a base pivotally attached to the vertically-oriented portion, a first connector supported by the base, and a first support portion and a second support portion supported at least in part by the first connector, the second platform portion being supported by the base, the first connector having a first end and an opposite second end, the first end of the first connector being pivotally attached relative to the base, and the first support portion and the second support portion each including a first end and an opposite second end, the first support portion and the second support portion being attached adjacent the second ends thereof to the second end of the first connector, and the first platform portion being supported by the linkage portion adjacent the first ends of the first support portion and the second support portion; the first platform portion including first patient support portions for supporting portions of the patient's head and upper torso; and the second platform portion including second patient support portions for supporting portions of the patient's legs; where the first platform portion is pivotal relative to the base via pivotal movement of the first connector; where the second platform portion is pivotal relative to the column portion via pivotal movement of the base; and where the pivotal movement of the first platform portion relative to the base, and pivotal movement of the second platform portion relative to the column portion serves to separately articulate the first platform portion and the second platform portion relative to one another.

(6) In yet another aspect, the present disclosure provides a surgical platform system facilitating manipulation of a patient supported thereby, the surgical platform system including a support supporting a linkage portion, a first support platform, and a second support platform relative to the ground, the linkage portion including a base pivotally attached to the support, a first connector supported by the base, and a first support portion and a second support portion supported at least in part by the first connector, the second platform portion being supported by the base, the first connector having a first end and an opposite second end, the first end of the first connector being pivotally attached relative to the base, and the first support portion and the second support portion each including a first end and an opposite second end, the first support portion and the second support portion being attached adjacent the second ends thereof to the second end of the first connector, and the first platform portion being supported by the linkage portion adjacent the first ends of the first support portion and the second support portion; the first platform portion including first patient support portions for supporting portions of the patient's head and upper torso; and the second platform portion including second patient support portions for supporting portions of the patient's legs; where the first platform portion is pivotal relative to the base via pivotal movement of the first connector; where the second platform portion is pivotal relative to the support via pivotal movement of the base; and where the pivotal movement of the first platform portion relative to the base, and pivotal movement of the second platform portion relative to the support serves to separately articulate the first platform portion and the second platform portion relative to one another.

(7) The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The techniques of this disclosure generally relate to a surgical platform system.
- (2) FIG. 1 is a top, side, perspective view of a surgical platform system of the present disclosure with a patient positioned thereon;
- (3) FIG. 2 is a top, side, perspective view of the surgical platform system of FIG. 1;
- (4) FIG. 3 is a first lateral side, elevational view of the surgical platform system of FIG. 1 with the patient positioned thereon;
- (5) FIG. 4 is a second lateral side, elevational view of the surgical platform system of FIG. 1 with the patient positioned thereon;
- (6) FIG. 5 is a front, elevational view of the surgical platform system of FIG. 1 with the patient positioned thereon;
- (7) FIG. 6 is a rear, elevational view of the surgical platform system of FIG. 1 with the patient positioned thereon;
- (8) FIG. 7 is a top, plan view of the surgical platform system of FIG. 1 with the patient positioned thereon;
- (9) FIG. 8 is a first lateral side, elevational view of the surgical platform system of FIG. 1 including portions thereof oriented to position the patient in a Trendelenburg position;
- (10) FIG. 9 is a first lateral side, elevational view of the surgical platform system of FIG. 1 including portions thereof oriented to position the patient in a Reverse-Trendelenburg position;
- (11) FIG. 10 is a side, top, perspective view of the surgical platform system of FIG. 1 being positioned relative to a gantry of a surgical robotic system or robotic surgical guidance system;
- (12) FIG. 11 is a side, top, perspective view of the surgical platform system of FIG. 1 interconnected relative to the gantry of FIG. 10;
- (13) FIG. 12 is a first lateral side, elevational view of the surgical platform system of FIG. 1 interconnected relative to the gantry of FIG. 10;
- (14) FIG. 13 is a side, top, perspective view of the surgical platform system of FIG. 1 interconnected relative to the gantry of FIG. 10 with portions of the surgical platform system and the gantry oriented to position the patient in a Trendelenburg position;
- (15) FIG. 14 is a side, top, perspective view of the surgical platform system of FIG. 1 interconnected relative to the gantry of FIG. 10 with portions of the surgical platform system and the gantry oriented to position the patient in a Reverse-Trendelenburg position;
- (16) FIG. 15 is a first lateral side, elevational view of the surgical platform system of FIG. 1 with the portions of the surgical platform and the gantry oriented to subject the patient's spine to extension;
- (17) FIG. 16 is a first lateral side, elevational view of the surgical platform system of FIG. 1 with the portions of the surgical platform and the gantry oriented to subject the patient's spine to flexion;
- (18) FIG. 17 is a second lateral side, elevational, fragmentary view of a portion of the surgical platform system of FIG. 1 depicting a platform portion thereof in a first position;
- (19) FIG. 18 is a second lateral side, elevational, fragmentary view of the portion of the surgical platform system of FIG. 1 depicting the platform portion thereof in a second position;
- (20) FIG. 19 is a second lateral side, elevational, fragmentary view of a portion of the surgical platform system of FIG. 1 depicting portions of another platform portion thereof in a first position;
- (21) FIG. 20 is a second lateral side, elevational, fragmentary view of a portion of the surgical platform system of FIG. 1 depicting portions of the another platform portion thereof in a second position;
- (22) FIG. 21 is a second lateral side, elevational, fragmentary view of a portion of the surgical platform system of FIG. 1 depicting portions of the platform portion and the patient positioned thereon;
- (23) FIG. 22 is a second lateral side, elevational view of an additional embodiment of a surgical

platform system according to the present disclosure;

(24) FIG. 23 is a second lateral side, elevational view of the surgical platform system of FIG. 22 with the patient positioned thereon including portions thereof oriented to subject the patient's spine to extension; and

(25) FIG. 24 is a second lateral side, elevational view of the surgical platform system of FIG. 22 with the patient positioned thereon including portions thereof oriented to subject the patient's spine to flexion.

DETAILED DESCRIPTION

(26) A preferred embodiment of a surgical platform system of the present disclosure is generally indicated by the numeral **10** in FIGS. 1-7. As depicted in FIG. 1, the surgical platform system **10** includes a first end E.sub.1, a second end E.sub.2, and a mid-longitudinal L.sub.1 extending through the first end E.sub.1 and the second end E.sub.2. The surgical platform system **10** includes a first platform portion **12**, a second platform portion **14**, a linkage portion **16**, and a support portion **18**. The first platform portion **12** and the second platform portion **14** can each support portions of a patient P thereon, the linkage portion **16** is used in adjusting the position and orientation of the first platform portion **12** and the second platform portion **14** relative to one another, and the support portion **18** supports the first platform portion **12**, the second platform portion **14**, and the linkage portion **16** relative to the ground. The first support platform **12** can be used in supporting the head and the torso of the patient P, the second platform portion **14** can be used in supporting the legs of the patient P, and the linkage portion **16** can be used to adjust the head and torso relative to the legs of the patient P by positioning and orienting the first support platform **12** and the second support platform **14** relative to one another. For example, the surgical platform system **10** can be used to adjust the first platform portion **12** and the second platform portion **14** to position/orient the patient in the Trendelenburg position (FIG. 8) and the Reverse-Trendelenburg position (FIG. 9).

(27) As discussed below, the surgical platform system **10** can be positioned and repositioned relative to, be interconnected with, and used in association with a surgical robotic system or robotic surgical guidance system (hereinafter referred to as "robotic system") generally indicated the letter R (FIGS. 10-16). And the robotic system R can be used for performing surgery or facilitating performance of surgery, and such surgery, for example, can included spinal surgery on the spine of the patient P.

(28) Ultimately, the surgical platform system **10** can be operatively attached to a gantry G of the robotic system R, and a robotic arm A supported by the gantry G can be used to perform surgery or facilitate performance of surgery on the patient P. In addition to the relative adjustment of the first platform portion **12** and the second platform portion **14** by the surgical platform system **10**, the gantry G, as discussed below, also can be used in adjusting portions relative to other portions of the surgical platform system **10**. The relative adjustment of the first platform portion **12** and the second platform portion **14** by the surgical platform system **10** and/or the adjustment afforded by the operative attachment of the surgical platform system **10** to the gantry G can be used to manipulate the body of the patient P and position/orient and reposition/reorient the patient P with respect to the robotic arm A.

(29) The surgical platform system **10** and/or the robotic system R can include a controller or controllers for controlling motorized actuators included in the surgical platform system **10** and/or the gantry G to facilitate the operation thereof. In some embodiments, for example, one of more controllers of the surgical platform system **10** and/or the robotic system R can coordinate movement therebetween by moving the first platform portion **12** and the second platform portion **14** relative to one another.

(30) During use, the support portion **18** can be used to facilitate movement of the first platform portion **12**, the second platform portion **14**, and the linkage portion **16**, and can be used positioning the first platform portion **12** relative to the robotic system R. The first platform portion **14** can then

be interconnected with the robotic system R or a sub-system (not shown) positioned relative to the robotic system. The support portion **18**, as depicted in FIGS. **1-4**, includes a horizontally-oriented portion **20** and a vertically-oriented portion **22**. The horizontally-oriented portion **20** is used in supporting the vertically-oriented portion **22** relative to the ground, and the vertically-oriented portion **22** is used in supporting the first platform portion **12**, the second platform portion **14**, and the linkage portion **16** relative to the horizontally-oriented portion **20**. After the support portion **18** is properly positioned relative to the robotic system R, the first platform portion **12** can be interconnected with the robotic system R or the sub-system positioned relative to the robotic system.

(31) As depicted in FIG. **1**, the horizontally-oriented portion **20** includes a first end member **24** at a first end **25** thereof (collocated with the first end E.sub.1), a second end member **26** at a second end **27** thereof (collocated with the second end E.sub.2), and a cross member **30** extending between the first end member **24** and the second end member **26**. The cross member **30** can be aligned with a mid-longitudinal axis L.sub.2 (FIG. **1**) of the horizontally-oriented portion **20**, can be used to connect the first end member **24** and the second end member **26**, and can be expandable and contractable to expand and contract a length of the horizontally-oriented portion **20** along the mid-longitudinal axis L.sub.2. Each of the first end member **24** and the second end member **26** includes an upper surface **32**, a lower surface **34**, a first lateral end **36**, and a second lateral end **38**. Furthermore, casters **40** can be attached relative to the lower surfaces **34** adjacent the first lateral ends **36** and the second lateral ends **38** of the of the first end member **24** and the second end member **26** to space the first end member **24** and the second end member **26** from the ground and to facilitate movement of the support portion **14**.

(32) The surgical platform system **10** initially can be positioned relative to the robotic system R using a positioner **42** having portions provided as part of the surgical platform system **10** and the robotic system R. To illustrate, the positioner **42** can include a receiver portion **44** (FIG. **5**) that can be provided as part of the support portion **18**, and a tongue portion **46** (FIG. **10**) that can be attached to or positioned relative to the robotic system R, or vice versa. A similar positioner that can be used as part of the surgical platform system **10** and the robotic system R is disclosed in U.S. Ser. No. 17/740,559, filed May 10, 2022, which is hereby incorporated by reference herein.

(33) The receiver portion **44** can be provided at and adjacent the first end **25**, can be formed by and/or attached relative to portions of the first end member **24** and the cross member **30**. The receiver portion **44** can be formed as a tunnel formed in portions of the first end member **24** and the cross member **30**. The receiver portion **44** includes a first sidewall portion **50** with portions provided adjacent a first lateral side **52** of the cross member **30**, a second sidewall portion **54** with portions provided adjacent a second lateral side **56** of the cross member **30**, and an upper wall portion **58** formed in part by the lower surface **34** of the first end member **24** and portions of the cross member **30**.

(34) Portions of the first sidewall portion **50**, the second sidewall portion **54**, and the upper wall portion **58** form the tunnel to define a receiving area A.sub.1 for receiving the tongue portion **46**. Furthermore, each of the first sidewall portion **50** and the second sidewall portion **54** can include various apertures **60** adjacent the receiving area A.sub.1 that are spaced therealong, and include various bumper wheels **62** rotatably mounted in the various apertures **60**. Portions of the bumper wheels **62** can extend into the receiving area A.sub.1. When the tongue portion **46** is received in the receiving area A.sub.1, the bumper wheels **62** are used to both guide and position the tongue **46** relative the first sidewall portion **50** and the second sidewall portion **54** (and the remainder of the horizontally-oriented portion **20**).

(35) The tongue portion **46** can be attached to (FIG. **10**) or otherwise positioned relative to the robotic system R, and includes a first end **64** and an opposite second end **66**. The tongue portion **46** can be attached to the robotic system R at and adjacent the first end **64**. Furthermore, the tongue portion **46** can include a first lateral side surface **70**, a second lateral side surface **72**, and a bottom

surface **74** extending between the first end **64** and the second end **66**. Adjustable feet **76** can be attached to the bottom surface **74** adjacent the second end **66** to facilitate leveling of at least portions of the tongue portion **46**.

(36) As depicted in FIG. **10**, to initially position the support portion **18** relative to the robotic system R, the horizontally-oriented portion **20** of the support portion **14** can be positioned so that the tongue portion **46** is received in the receiver portion **44**. In doing so, the tongue portion **46** is inserted between the first sidewall portion **50** and the second sidewall portion **54**, under the upper wall portion **58**, and into the receiving area A.sub.1. As the tongue portion **46** is received in the receiving area A.sub.1, the first lateral side surface **70** contacts the bumper wheels **62** rotatably mounted to the first sidewall portion **50**, and the second lateral side surface **72** contacts the bumper wheels **62** rotatably mounted to the second sidewall portion **54**. Such contact affords relative movement of the tongue portion **46** to the receiver portion **44** that guides the tongue portion **46** into the receiving area A.sub.1 to initially position the surgical platform system **10** relative to the robotic system R.

(37) When the surgical platform system **10** is attached relative to the robotic system R, adjustment of the first platform portion **12** and the second platform portion **14** relative to one another (via operation of the surgical platform system **10** and/or the robotic system R) can change the distance between the second platform portion **14** and the gantry G, and the position of the tongue portion **46** in the receiving area A1 can correspondingly change during such operation. As such, the tongue portion **46** can be moved into and out of the receiving A1 according to the changes in the distance between the second platform portion **14** and the gantry G.

(38) As depicted in FIG. **2-4**, the vertically-oriented portion **22** includes a telescoping column **80** for positioning/orienting and repositioning/reorienting the first platform portion **12**, the second platform portion **14** and the linkage portion **16** relative to the horizontally-oriented portion **20**. The telescoping column **80** includes a lower portion **82** and an upper portion **84**. The lower portion of the telescoping column **80** is supported by the cross member **30**, and the upper portion **84** can be telescopically moved upwardly and downwardly relative to the lower portion **82**. The telescopic expansion and contraction of the telescoping column **80** can be used to correspondingly raise and lower the first platform portion **12**, the second platform portion **14**, and the linkage portion **16** relative to the horizontally-oriented portion **20** to position/orient and reposition/reorient the first platform portion **12**, the second platform portion **14**, and the linkage portion **16** between a lower position and an upper position.

(39) The linkage portion **16**, as depicted in FIGS. **2-4** and **6**, includes a base formed by a first base portion **90** positioned adjacent a first side of the telescoping column **80** and a second base portion **92** positioned adjacent a second side of the telescoping column **80**. As discussed below, the first base portion **90** and the second base portion **92** are pivotally attached to a portion of the upper portion **84**, and the base formed by the first base portion **90** and the second base portion **92** affords pivotal movement of other portions of the linkage portion **16** relative thereto. The first base portion **90** and the second base portion **92** can each include a lower portion **94** and an upper portion **96**. Furthermore, as depicted in FIG. **5**, the upper portion **84** of the telescoping column **80** can include an extended portion **100** with a first post **102** on a first side of the extended portion **100**, and a second post **104** on a second side of the extended portion **100**. The lower portion **94** of the first base portion **90** can be pivotally attached to the first post **102**, and the lower portion **94** of the second base portion **92** can be pivotally attached to the second post **104**. The lower portions **94** of the first base portion **90** and the second base portion **92** are pivotally attached to the first post **102** and the second post **104**, respectively, adjacent ends of the lower portions **94** closest to the first end E.sub.1. As discussed below, pivotal movement of the first base portion **90** and the second base portion **92** affords articulation of the first platform portion **12** and the second platform portion **14** relative to the telescoping column **80**.

(40) In addition to being attached relative to one another by being pivotally attached to the first

post **102** and the second post **104**, the first base portion **90** and the second base portion **92** can be attached to one another via a first connector **106** and a second connector **108**. The first connector **106** and the second connector **108** extend between and are fastened to the first base portion **90** and the second base portion **92**. As depicted in FIGS. **5** and **6**, the first connector **106** is attached to the lower portions **94** of the first base portion **90** and the second base portion **92** above where the first base portion **90** and the second base portion **92** are pivotally attached to the first post **102** and the second post **104**, respectively; and the second connector **108** is attached to the lower portions **94** of the first base portion **90** and the second base portion **92** adjacent ends of the lower portions **94** closest to the second end E.sub.2.

(41) The attachment to the first post **102** and the second post **104**, and, as discussed below, the first connector **106** and the second connector **108** serve in spacing the first base portion **90** and the second base portion **92** apart from one another and providing a rigid and stable connection therebetween. For example, as depicted in FIG. **5**, the first connector **106** can be formed as a single rod portion extending between and attaching the first base portion **90** and the second base portion **92** to one another. And, as depicted in FIG. **6**, the second connector **108** can include a web portion **110**, a first rod portion **112** on a first side of the web portion **110**, and a second rod portion **114** on a second side of the web portion **110**. Portions of a lower portion of the web portion **110**, the first rod portion **112**, and the second rod portion **114** extend between and attach the first base portion **90** and the second base portion **92** to one another. The first connector **106** can be fixedly attached between the first base portion **90** and the second base portion **92**, and the portions the web portion **110**, the first rod portion **112**, and the second rod portion **114** of the second connector **108** can be pivotally connected to the first base portion **90** and the second base portion **92**.

(42) As depicted in FIG. **6**, the upper portions of the first base portion **90** and the second base portion **92** can each include a recess **120** for receiving portions of the second platform portion **14** to facilitate attachment of the second platform portion **14** thereto. The second support platform **14**, as depicted in FIG. **2**, includes a first support portion **122** and a second support portion **124**, and the first support portion **122** and the second support portion **124** each include a first end **126** and a second end **128**. The first support portion **122** and the second support portion can each be formed, for example, as beams, rails, rods, or tubes having lengths extending between the first end **126** and the second end **128**. Portions of the first support portion **122** between the first end **126** and the second end **128** are attached to the first base portion **90** via receipt and attachment thereof in the recess **120**, and portions of the second support portion **124** between the first end **126** and the second end **128** are attached to the second base portion **92** via receipt and attachment thereof in the recess **120**.

(43) A first upper thigh support **130**, a second upper thigh support **132**, a first lower thigh support **134**, a second lower thigh support **136**, a support plate **140**, and a lower leg support **142** are supported by and/or between the first support portion **122** and/or the second support portion **124**. As depicted in FIG. **2**, the first upper thigh support **130** and the first lower thigh support **134** are supported by the first support portion **122**. The second upper thigh support **132** and the second lower thigh support **136** are supported by the second support portion **124**. The support plate **140** (supporting the lower leg support **142**) is supported between the first support portion **122** and the second support portion **124**. The first upper thigh support **130**, the second upper thigh support **132**, the first lower thigh support **134**, the second lower thigh support **136**, and the support plate **140** can be configured similarly to those disclosed in U.S. Ser. No. 17/704,759, filed Mar. 25, 2022, which is hereby incorporated by reference herein, and the positions thereof can be adjusted relative to the first support portion **122** and the second support portion **124** to accommodate patients of different sizes.

(44) The second platform portion **14** can be pivoted relative to the telescoping column **80** (and the support portion **18**) using a first actuator **144** that is actuatable (via expansion and contraction thereof) between at least a first position (FIG. **17**) and a second position (FIG. **18**). The first

actuator **144** has a first end **146** pivotally attached to the horizontally-oriented portion **20**, and a second end **148** pivotally attached to the second connector **108** (between the first rod portion **112** and the second rod portion **114**, and portions of the web portion **110**). The first end **146**, for example, can be pivotally attached to the horizontally-oriented portion **20** in an aperture formed therein or a clevis extending outwardly therefrom. Actuation of the first actuator **144** moves the second connector **108** toward or away from the horizontally-oriented portion **20**, and correspondingly pivots the first base portion **90**, the second base portion **92**, and the second platform portion **14** attached thereto relative to the telescoping column **80** (and the support portion **18**). In doing so, the first ends **126** of the first support portion **122** and the second support portion **124** are tipped downwardly via expansion of the first actuator **144** or tipped upwardly via contraction of the first actuator **144** to correspondingly tilt the second platform portion **14**.

(45) The linkage portion **16** includes a third support portion **150** and a fourth support portion **152** supporting the first platform portion **12**, and the third support portion **150** and the fourth support portion **152** are pivotally attached to the second connector **108**. The third support portion **150** and the fourth support portion **152** can each be formed, for example, as beams, rails, rods, or tubes having lengths extending between opposite ends thereof. The first platform portion **12** can be oriented relative to the column **80** (and the support portion **18**) via pivotal movement of the first base portion **90** and the second base portion **92**, via pivotal movement of the second connector **108** relative to the first base portion **90** and the second base portion **92**, and via pivotal movement of the third support portion **150** and the fourth support portion **152** relative to the second connector **108**. Pivotal movement of the second connector **108** relative to the first base portion **90** and the second base portion **92**, and pivotal movement of the third support portion **150** and the fourth support portion **152** relative to the second connector **108** also orients the first platform portion **12** relative to the second platform portion **14**.

(46) As depicted in FIG. 2, the third support portion **150** and the fourth support portion **152** each include a first end **154** and a second end **156**. The first platform portion **12** is supported by the third support portion **150** and the fourth support portion **152** adjacent the first ends **154** thereof. And the second ends **156** of the third support portion **150** and the fourth support portion **152** are spaced apart by and pivotally attached to an upper portion of the web portion **110** of the second connector **108**. The third support portion **150** and the fourth support portion **152** also are spaced apart by an end portion **158** of the linkage portion **16** at the first ends **154** thereof, and by a third connector **160** intermediate the first ends **154** and the second ends **156**.

(47) As depicted in FIGS. 19-21, pivotal movement of the second connector **108** relative to first base portion **90** and the second base portion **92** is afforded by use of a second actuator **162**. The second actuator **162**, as depicted in FIG. 21, has a first end **164** pivotally attached to the first connector **106**, and a second end **166** pivotally attached to the upper portion of the web portion **110** of the second connector **108**. The second actuator **162** is actuatable (via expansion and contraction thereof) between at least a first position (FIG. 19) and a second position (FIG. 20). Given the pivotal connection of the lower portion of the second connector **108** to the first base portion **90** and the second base portion **92**, actuation of the second actuator **162** pivots the second connector **108** relative to the first base portion **90** and the second base portion **92** to correspondingly move the first ends **154** of the third support portion **150** and the fourth support portion **152** toward (via contraction) or away (via expansion) from the first end E.sub.1. Thus, such pivotal movement serves in orienting the first platform portion **12** relative to the telescoping column **80** (and the support portion **18**) and to the second support portion **14** by moving the first platform portion toward or away from the first end E.sub.1.

(48) As also depicted in FIGS. 19-21, pivotal movement of the third support portion **150** and the fourth support portion **152** relative to the second connector **108** is afforded by use of a third actuator **170**. The third actuator **170** has a first end **172** pivotally attached to the horizontally-oriented portion **20**, and a second end **174** pivotally attached to the third connector **160**. The first

end **172**, for example, can be pivotally attached to the horizontally-oriented portion **20** in an aperture formed therein or a clevis extending outwardly therefrom. The third actuator **170** is actuatable (via expansion and contraction thereof) between at least a first position (FIG. **19**) and a second position (FIG. **20**). Given the pivotal connection of the third support portion **150** and the fourth support portion **152** to the upper portion of the second connector **108**, actuation of the third actuator **170** pivots the third support portion **150** and the fourth support portion **152** relative to the second connector **108** to correspondingly tip the first ends **154** of the third support portion **150** and the fourth support portion **152** downwardly (via contraction) or upwardly (via expansion). Thus, such pivotal movement serves in orienting the first platform portion **12** relative to the telescoping column **80** (and the support portion **18**) and to the second support portion **14** by moving the first platform portion **12** downwardly or upwardly.

(49) The pivotal movement of the second connector **108** relative to the first base portion **90** and the second base portion **92**, and the pivotal movement of the third support portion **150** and the fourth support portion **152** relative to the second connector **108** orients the first platform portion **12** relative to the second platform portion **14**. As such, the first platform portion **12** and the second platform portion **14** are separately articulatable via operation of the second actuator **162** and the third actuator **170**, and, when the patient P is supported on the first platform portion **12** and the second platform portion **14**, such separate articulation affords subjecting the patient's body to flexion and extension.

(50) As discussed below, actuation of the first actuator **144**, the second actuator **162**, and the third actuator **170** can be automated and controlled by the controller or controllers of the surgical platform system **10** and/or the robotic system R. To illustrate, the controller or controllers can be used, for example, to orient the first platform portion **12** and the second platform portion **14** relative to one another in the above-described Trendelenburg position (FIG. **8**), Reverse-Trendelenburg position (FIG. **9**), or other below-described orientations. As discussed below, the controller or controllers also can be used in conjunction with the robotic system R to adjust the position of the first platform portion **12** and the second platform portion **14** relative to one another.

(51) As depicted in FIG. **2**, the first platform portion **12** includes an first end **180**, an opposite second end **182**, a first end portion **184** provided at and adjacent the first end **180**, a second end portion **186** at and adjacent the second end **182**, a fifth support portion **190** extending between the first end portion **184** and the second end portion **186**, and a sixth support portion **192** extending between the first end portion **184** and the second end portion **186**. The fifth support portion **190** and the sixth support portion **192** can each be formed, for example, as beams, rails, rods, or tubes having lengths extending between opposite ends thereof. The first end portion **184** can include an engagement portion **194** similar to that disclosed in U.S. Ser. No. 17/704,759, filed Mar. 25, 2022, for interfacing with portions of the robotic system R to facilitate interconnection of the first platform portion **12** with the robotic system R or the sub-system positioned relative to the robotic system.

(52) Additionally, the first platform portion **12** can include a head support **196** and a chest support **198** similar to those also disclosed in U.S. Ser. No. 17/704,759, filed Mar. 25, 2022. The head support **196** and the chest support **198** are supported by and/or between the fifth support portion **190** and the sixth support portion **192**, and the positions thereof can be adjusted relative to the fifth support portion **190** and the sixth support portion **192** to accommodate patients of different sizes. Furthermore, in addition to the head support **196** and the chest support **198**, the first platform portion **12** can include a first arm support **200** supported by the fifth support portion **190** and a second arm support **202** supported by the sixth support portion **192**.

(53) The first platform portion **12** can be pivotally attached relative to the linkage portion **16**. As depicted in FIGS. **3** and **4**, the second end portion **186** can include a first extension portion **204** and a second extension portion **206** extending downwardly therefrom. The first extension portion **204** is provided adjacent the fifth support portion **190** and the second extension portion **206** is provided

adjacent the sixth support portion **192**. Each of the first extension portion **204** and the second extension portion **206** include a first leg portion **210** attached to the second end portion **186**, and a second leg portion **212** extending outwardly from the first leg portion **210**. First leg portions **210** can each be pivotally attached to the linkage portion **16** via receipt of pins **214** extending therethrough and into the end portion **158** of the linkage portion **16**. And the first platform portion **12** can be fixed in a first pivotal position relative to the linkage portion **16** via receipt of removable pins **216** extending through the second leg portions **210** and into the end portion **158** of the linkage portion **16**. When the removable pins **216** are removed, the first platform portion **12** can pivot relative to the linkage portion **16** (and the second platform portion **12**) into various other pivotal positions. As discussed below, pivot movement afforded by the pivotal attachment of the first platform portion **12** relative to the linkage portion **16** allows the surgical gantry **G** to further articulate the position of the first platform portion **12** relative to the second platform portion **14**.

(54) After the surgical platform system **10** is initially positioned relative to the robotic system **R** (FIG. **10**), and thereafter, the tongue portion **46** is received in the receiving area A.sub.1, the engagement portion **194** can be engaged to a complimentary engagement portion **220** similar to that disclosure in U.S. Ser. No. 17/704,759, filed Mar. 25, 2022, provided on the robotic system **R** or the sub-system positioned relative to the robotic system **R**. As depicted in FIG. **10**, the complimentary engagement portion **220** is provided on the gantry **G**. With surgical platform system **10** interconnected (FIGS. **11-16**) with the gantry **G** via engagement of the engagement portion **194** with the complimentary engagement portion **220**, the surgical platform portion **10** can work together with portions of the gantry **10** to articulate the first platform portion **12** and the second platform portion **14** relative to position and orient the patient **P** relative to the robotic system **R**.

(55) As depicted in FIG. **10-16**, the gantry **G** can include a platform manipulator **222** supporting the complimentary engagement portion **220** and positioned above the tongue portion **46**. The platform manipulator **222** can be used in raising, lowering, and tilting (upwardly or downwardly) portions of the first platform portion **12** relative to the second platform portion **14**. To illustrate, when the surgical platform system **10** is interconnected with the gantry **G**, and the first platform portion **12** is in the first pivotal position relative to the linkage portion, the surgical platform system **10** and the platform manipulator **222** can be operated in a coordinated fashion to position/orient the first platform portion **12** and the second platform portion **14** so that the patient **P** is moved into the Trendelenburg position (FIG. **13**) and the Reverse-Trendelenburg position (FIG. **14**). In doing so, the platform manipulator **222** can be used in tilting the first platform portion **12** adjacent the first end **180** downwardly from the Trendelenburg position or upwardly for the Reverse-Trendelenburg position. When the removable pins **216** are removed to allow the first platform portion **12** to pivot relative to the linkage portion **16**, the surgical platform system **10** and the platform manipulator **222** can be operated in a coordinated fashion (via use of the controller or controllers of the surgical platform system **10** and/or the robotic system **R**) to position/orient the first platform portion **12** and the second platform portion **14** so that the patient **P** is subjected to extension (FIG. **15**) or flexion (FIG. **16**). In doing so, the platform manipulator **222** can be used in tilting the first platform portion **12** adjacent the first end **180** upwardly to facilitate subjecting the patient's spine to extension or downwardly to facilitate subjecting the patient's spine to flexion. Thus, operation of the surgical platform system **10** and the platform manipulator **222** can be used to position/orient and reposition/reorient the patient's body with respect to the robotic arm **A**, and the robotic arm **A** can be used for performing surgery or facilitating performance of surgery on the patient **P**.

(56) Another embodiment of the surgical platform system **10'** can be configured to facilitate pivoting of a first platform portion **12'** relative to a linkage portion **16'**. As depicted in FIGS. **22-24**, the surgical platform system **10'** includes similar features to the surgical platform system **10**, and like numerals are used to in describing these similar features. Unlike the surgical platform system **10**, the surgical platform system **10'** includes a modified end portion **158'** of the linkage portion **16**, a modified first end portion **184'** of the first platform portion **12'**, and a fourth actuator **224**

extending between the modified end portion **158'** and the modified first end portion **184'**.
(57) Like the first platform portion **12** and the linkage portion **16**, the first platform portion **12** is pivotally attached to the linkage portion **16**. And the fourth actuator **224** is actuatable (via expansion and contraction thereof) between at least a first position (FIG. **23**) and a second position (FIG. **24**) to pivot the first platform portion **12'** relative to the linkage portion **16'**. The fourth actuator **224** has a first end **226** pivotally attached to the modified end portion **158'**, and a second end **228** pivotally attached to the modified first end portion **184'**. The modified end portion **158'** and the modified first end portion **184'** can each include a clevis **230** facilitating pivotal attachment, respectively, of the first end **226** and the second end **228**. Actuation of the fourth actuator **224** pivots the first platform portion **12'** relative to the linkage portion **16'** and tips the first end **180** of the first platform portion **12'** upwardly (FIG. **23**) or downwardly (FIG. **24**). Such upward and downward movement of the first end **180** of the first platform portion **12'** along with coordinated movement of other portions of the surgical platform **10'** can be used to subject the patient's spine to extension (FIG. **23**) and flexion (FIG. **24**). The operation of the fourth actuator **224** can be coordinated with operation of the first actuator **144**, the second actuator **162**, and/or the third actuator **170** using the controller or controllers of the surgical platform system **10** and/or the robotic system **R**.

(58) It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and the accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes of methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain aspect of this disclosure are described as being performed by a single module or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or modules associated with, for example, a medical device.

Claims

1. A surgical platform system facilitating manipulation of a patient supported thereby, the surgical platform system comprising: a support portion including a horizontally-oriented portion and a vertically-oriented portion, the horizontally-oriented portion having a first end, an opposite second end, a first end portion at the first end, a second end portion at the second end, and a cross member extending between the first end portion and the second end portion, and the vertically-oriented portion including a column portion supported by the horizontally-oriented portion, the column portion spacing a first platform portion, a second platform portion, and a linkage portion of the surgical platform system from the horizontally-oriented portion, the linkage portion including a first base portion positioned on a first side of the column portion, a second base portion positioned on a second side of the column portion, a first connector extending between the first base portion and the second base portion, and a first support portion and a second support portion supported at least in part by the first connector, the first base portion and the second base portion being pivotally attached relative to the column portion, and the second platform portion being supported by the first base portion and the second base portion, the first connector having a first end and an opposite second end, the first end of the first connector being pivotally attached relative to the first base portion and the second base portion, and the first support portion and the second support portion each including a first end and an opposite second end, the first support portion and the second support portion being attached adjacent the second ends thereof to the second end of the first connector, and the first platform portion being supported by the linkage portion adjacent the first ends of the first support portion and the second support portion; the first platform portion including first patient support portions for supporting portions of the patient's head and upper torso; and the

second platform portion including second patient support portions for supporting portions of the patient's legs; wherein the first platform portion is pivotal relative to the first base portion and the second base portion via pivotal movement of the first connector; wherein the second platform portion is pivotal relative to the column portion via pivotal movement of the first base portion and the second base portion; and wherein pivotal movement of the first platform portion relative to the first base portion and the second base portion and pivotal movement of the second platform portion relative to the column portion serve to separately articulate the first platform portion and the second platform portion relative to one another.

2. The surgical platform system of claim 1, further comprising a first actuator extending between the horizontally-oriented portion and the first connector, the first actuator being actuatable to expand and contract to pivot the first base portion, the second base portion, and the second platform portion relative to the column portion.

3. The surgical platform system of claim 1, wherein the second ends of the first support portion and the second support portion are pivotally attached to the second end of the first connector.

4. The surgical platform system of claim 3, further comprising a first actuator extending between the horizontally-oriented portion and a second connector attached between the first support portion and the second support portion, the first actuator being actuatable to expand and contract to pivot the first support portion and the second support portion relative to the first connector.

5. The surgical platform system of claim 4, further comprising a second actuator extending between the second end of the first connector and a third connector attached between the first base portion and the second base portion, the second actuator being actuatable to expand and contract to pivot the first connector relative to the first base portion and the second base portion.

6. The surgical platform system of claim 1, further comprising a first actuator extending between the horizontally-oriented portion and the first connector, a second actuator extending between the horizontally-oriented portion and a second connector attached between the first support portion and the second support portion, and a third actuator extending between the second end of the first connector and a third connector attached between the first base portion and the second base portion, the first actuator being actuatable to expand and contract to pivot the first base portion, the second base portion, and the second platform portion relative to the column portion, the second actuator being actuatable to expand and contract to pivot the first support portion and the second support portion relative to the first connector; and the third actuator being actuatable to expand and contract to pivot the first connector relative to the first base portion and the second base portion.

7. The surgical platform system of claim 6 in combination with a robotic system, further comprising: the robotic system including a gantry having a platform manipulator and an engagement portion supported by the platform manipulator; and the first platform portion including a first end and an opposite second end, the first end of the first platform portion including a complimentary engagement portion for engaging the engagement portion of the gantry, and the second end of the first platform portion being attached to the first ends of the first support portion and the second support portion; wherein, when the engagement portion and the complimentary engagement portion are engaged to one another, the surgical platform system is interconnected with the gantry, and the first actuator, the second actuator, the third actuator, and the platform manipulator can be actuated to position/orient the patient supported by the first platform portion and the second platform portion relative to the robotic system.

8. The combination of the surgical platform system and the robotic system of claim 7, wherein the second end of the first platform portion is pivotally attached to the first ends of the first support portion and the second support portion, and wherein the first actuator, the second actuator, the third actuator, and the platform manipulator can be actuated to pivot the first platform portion and the second platform portion relative to one another.

9. The surgical platform system of claim 1, wherein the column portion is telescopically expandable and contractable between a contracted first position and an expanded second position to

raise and lower the first platform portion, the second platform portion, and the linkage portion relative to the horizontally-oriented portion.

10. A surgical platform system facilitating manipulation of a patient supported thereby, the surgical platform system comprising: a horizontally-oriented portion and a vertically-oriented portion, the horizontally-oriented portion supporting the vertically-oriented portion, and the vertically-oriented portion including a column portion spacing a first platform portion, a second platform portion, and a linkage portion of the surgical platform system from the horizontally-oriented portion, the linkage portion including a base pivotally attached to the vertically-oriented portion, a first connector supported by the base, and a first support portion and a second support portion supported at least in part by the first connector, the second platform portion being supported by the base, the first connector having a first end and an opposite second end, the first end of the first connector being pivotally attached relative to the base, and the first support portion and the second support portion each including a first end and an opposite second end, the first support portion and the second support portion being attached adjacent the second ends thereof to the second end of the first connector, and the first platform portion being supported by the linkage portion adjacent the first ends of the first support portion and the second support portion; the first platform portion including first patient support portions for supporting portions of the patient's head and upper torso; and the second platform portion including second patient support portions for supporting portions of the patient's legs; wherein the first platform portion is pivotal relative to the base via pivotal movement of the first connector; wherein the second platform portion is pivotal relative to the column portion via pivotal movement of the base; and wherein pivotal movement of the first platform portion relative to the base and pivotal movement of the second platform portion relative to the column portion serve to separately articulate the first platform portion and the second platform portion relative to one another.

11. The surgical platform system of claim 10, further comprising a first actuator extending between the horizontally-oriented portion and the first connector, the first actuator being actuatable to expand and contract to pivot the base and the second platform portion relative to the column portion.

12. The surgical platform system of claim 10, wherein the second ends of the first support portion and the second support portion are pivotally attached to the second end of the first connector.

13. The surgical platform system of claim 12, further comprising a first actuator extending between the horizontally-oriented portion and a second connector attached between the first support portion and the second support portion, the first actuator being actuatable to expand and contract to pivot the first support portion and the second support portion relative to the first connector.

14. The surgical platform system of claim 13, further comprising a second actuator extending between the second end of the first connector and a third connector attached between portions of the base, the second actuator being actuatable to expand and contract to pivot the first connector relative to the base.

15. The surgical platform system of claim 10, further comprising a first actuator extending between the horizontally-oriented portion and the first connector, a second actuator extending between the horizontally-oriented portion and a second connector attached between the first support portion and the second support portion, and a third actuator extending between the second end of the first connector and a third connector attached between portions of the base, the first actuator being actuatable to expand and contract to pivot the base and the second platform portion relative to the column portion, the second actuator being actuatable to expand and contract to pivot the first support portion and the second support portion relative to the first connector; and the third actuator being actuatable to expand and contract to pivot the first connector relative to the base.

16. The surgical platform system of claim 15 in combination with a robotic system, further comprising: the robotic system including a gantry having a platform manipulator and an engagement portion supported by the platform manipulator; and the first platform portion including

a first end and an opposite second end, the first end of the first platform portion including a complimentary engagement portion for engaging the engagement portion of the gantry, and the second end of the first platform portion being attached to the first ends of the first support portion and the second support portion; wherein, when the engagement portion and the complimentary engagement portion are engaged to one another, the surgical platform system is interconnected with the gantry, and the first actuator, the second actuator, the third actuator, and the platform manipulator can be actuated to position/orient the patient supported by the first platform portion and the second platform portion relative to the robotic system.

17. The combination of the surgical platform system and the robotic system of claim 16, wherein the second end of the first platform portion is pivotally attached to the first ends of the first support portion and the second support portion, and wherein the first actuator, the second actuator, the third actuator, and the platform manipulator can be actuated to pivot the first platform portion and the second platform portion relative to one another.

18. The surgical platform system of claim 10, wherein vertically-oriented portion includes a column portion, and the column portion is telescopically expandable and contractable between a contracted first position and an expanded second position to raise and lower the first platform portion, the second platform portion, and the linkage portion relative to the horizontally-oriented portion.

19. A surgical platform system facilitating manipulation of a patient supported thereby, the surgical platform system comprising: a support supporting a linkage portion, a first support platform, and a second support platform relative to the ground, the linkage portion including a base pivotally attached to the support, a first connector supported by the base, and a first support portion and a second support portion supported at least in part by the first connector, the second platform portion being supported by the base, the first connector having a first end and an opposite second end, the first end of the first connector being pivotally attached relative to the base, and the first support portion and the second support portion each including a first end and an opposite second end, the first support portion and the second support portion being attached adjacent the second ends thereof to the second end of the first connector, and the first platform portion being supported by the linkage portion adjacent the first ends of the first support portion and the second support portion; the first platform portion including first patient support portions for supporting portions of the patient's head and upper torso; and the second platform portion including second patient support portions for supporting portions of the patient's legs; wherein the first platform portion is pivotal relative to the base via pivotal movement of the first connector; wherein the second platform portion is pivotal relative to the support via pivotal movement of the base; and wherein pivotal movement of the first platform portion relative to the base and pivotal movement of the second platform portion relative to the support serve to separately articulate the first platform portion and the second platform portion relative to one another.

20. The surgical platform system of claim 19 in combination with a robotic system, further comprising: the robotic system including a gantry having a platform manipulator and an engagement portion supported by the platform manipulator; and the first platform portion including a first end and an opposite second end, the first end of the first platform portion including a complimentary engagement portion for engaging the engagement portion of the gantry, and the second end of the first platform portion being attached to the first ends of the first support portion and the second support portion; wherein, when the engagement portion and the complimentary engagement portion are engaged to one another, the surgical platform system is interconnected with the gantry, and the first platform portion and the second platform portion are manipulatable relative to one another to position/orient the patient supported by the first platform portion and the second platform portion relative to the robotic system.
